# SG No. 66 $D_{2h}^{20}$ $Cccm$ [ orthorhombic ]

* plus set: $+[0,\ 0,\ 0]$, $+\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$

Table 1: Wyckoff site: `4a`, site symmetry: `222`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ 0,\ \frac{1}{4}\right]$ | `[1,2,3,4]` |
| 2 | $\left[0,\ 0,\ \frac{3}{4}\right]$ | `[5,6,7,8]` |

Table 2: Wyckoff site: `4b`, site symmetry: `222`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$ | `[1,2,3,4]` |
| 2 | $\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$ | `[5,6,7,8]` |

Table 3: Wyckoff site: `4c`, site symmetry: `..2/m`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ 0,\ 0]$ | `[1,2,5,6]` |
| 2 | $\left[0,\ 0,\ \frac{1}{2}\right]$ | `[3,4,7,8]` |

Table 4: Wyckoff site: `4d`, site symmetry: `..2/m`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ \frac{1}{2},\ 0\right]$ | `[1,2,5,6]` |
| 2 | $\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$ | `[3,4,7,8]` |

Table 5: Wyckoff site: `4e`, site symmetry: `..2/m`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4},\ \frac{1}{4},\ 0\right]$ | `[1,2,5,6]` |
| 2 | $\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}\right]$ | `[3,4,7,8]` |

Table 6: Wyckoff site: `4f`, site symmetry: `..2/m`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4},\ \frac{3}{4},\ 0\right]$ | `[1,2,5,6]` |
| 2 | $\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{2}\right]$ | `[3,4,7,8]` |

Table 7: Wyckoff site: `8g`, site symmetry: `2..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ 0,\ \frac{1}{4}\right]$ | `[1,4]` |

*continued ...*

Table 7

| No. | position | mapping |
|---|---|---|
| 2 | $\left[-x,\ 0,\ \frac{1}{4}\right]$ | `[2,3]` |
| 3 | $\left[-x,\ 0,\ \frac{3}{4}\right]$ | `[5,8]` |
| 4 | $\left[x,\ 0,\ \frac{3}{4}\right]$ | `[6,7]` |

Table 8: Wyckoff site: `8h`, site symmetry: `.2.`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ y,\ \frac{1}{4}\right]$ | `[1,3]` |
| 2 | $\left[0,\ -y,\ \frac{1}{4}\right]$ | `[2,4]` |
| 3 | $\left[0,\ -y,\ \frac{3}{4}\right]$ | `[5,7]` |
| 4 | $\left[0,\ y,\ \frac{3}{4}\right]$ | `[6,8]` |

Table 9: Wyckoff site: `8i`, site symmetry: `..2`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ 0,\ z]$ | `[1,2]` |
| 2 | $\left[0,\ 0,\ \frac{1}{2}-z\right]$ | `[3,4]` |
| 3 | $[0,\ 0,\ -z]$ | `[5,6]` |
| 4 | $\left[0,\ 0,\ z+\frac{1}{2}\right]$ | `[7,8]` |

Table 10: Wyckoff site: `8j`, site symmetry: `..2`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ \frac{1}{2},\ z\right]$ | `[1,2]` |
| 2 | $\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$ | `[3,4]` |
| 3 | $\left[0,\ \frac{1}{2},\ -z\right]$ | `[5,6]` |
| 4 | $\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$ | `[7,8]` |

Table 11: Wyckoff site: `8k`, site symmetry: `..2`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4},\ \frac{1}{4},\ z\right]$ | `[1,2]` |
| 2 | $\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$ | `[3,4]` |
| 3 | $\left[\frac{3}{4},\ \frac{3}{4},\ -z\right]$ | `[5,6]` |
| 4 | $\left[\frac{1}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$ | `[7,8]` |

Table 12: Wyckoff site: `8l`, site symmetry: `..m`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ 0]$ | `[1,6]` |
| 2 | $[-x,\ -y,\ 0]$ | `[2,5]` |

*continued ...*

Table 12

| No. | position | mapping |
|---|---|---|
| 3 | $\left[-x,\, y,\, \frac{1}{2}\right]$ | `[3,8]` |
| 4 | $\left[x,\, -y,\, \frac{1}{2}\right]$ | `[4,7]` |

Table 13: Wyckoff site: `16m`, site symmetry: `1`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\, y,\, z]$ | `[1]` |
| 2 | $[-x,\, -y,\, z]$ | `[2]` |
| 3 | $\left[-x,\, y,\, \frac{1}{2}-z\right]$ | `[3]` |
| 4 | $\left[x,\, -y,\, \frac{1}{2}-z\right]$ | `[4]` |
| 5 | $[-x,\, -y,\, -z]$ | `[5]` |
| 6 | $[x,\, y,\, -z]$ | `[6]` |
| 7 | $\left[x,\, -y,\, z+\frac{1}{2}\right]$ | `[7]` |
| 8 | $\left[-x,\, y,\, z+\frac{1}{2}\right]$ | `[8]` |